Systems Architecture

Integer Numbers

Characters, Words, and Bytes

1 bit (2^0)	bit	*Bi*nary Dig*it*
4 bits (2^2)	nibble	Half a byte
8 bits (2^3)	byte	Minimum address unit
16 bits (2^4)	half word	Two bytes / Half a word
32 bits (2^5)	word	Size of internal registers
64 bits (2^6)	double word	Two Words
128 bits (2^7)	extended word	Quad (four) Words or *Paragraph* (Intel)

numeric quantity	Integer / Floating-point (real) value
picture element	Colour of a picture element (pixel)
character	A text character (ASCII or Unicode)
instruction	A machine instruction

Number Bases

Bin	Binary	Base 2	1-bit
Oct	Octal	Base 8	3-bit
Dec	Decimal	Base 10	3.3-bit
Hex	Hexadecimal	Base 16	4-bit

	$base^3$	$base^2$	$base^1$	$base^0$
Bin	8	4	2	1
Oct	512	64	8	1
Dec	1000	100	10	1
Hex	4096	256	16	1

Equivalents Table

Bin	Oct	Dec	Hex	Bin	Oct	Dec	Hex
0000	00	00	0	1000	10	08	8
0001	01	01	1	1001	11	09	9
0010	02	02	2	1010	12	10	A
0011	03	03	3	1011	13	11	B
0100	04	04	4	1100	14	12	C
0101	05	05	5	1101	15	13	D
0110	06	06	6	1110	16	14	E
0111	07	07	7	1111	17	15	F

$1100_2 = 14_8 = 12_{10} = C_{16}$

Hardware Control

Bit	Component	Status	Value
0	Heater 1	off	0
1	Heater 2	on	1
2	Heater 3	off	0
3	Valve 1	on	1
4	Valve 2	off	0
5	Valve 3	off	0
6	Pump 1	on	1
7	Pump 2	on	1

$11\ 001\ 010_2 \Rightarrow 1100\ 1010_2 \Rightarrow 11001010_2$

$312_8 = CA_{16} = 202_{10}$

$(12 \times 16) + 10$

Decimal to Base n

Repeat

Divide number by *base*

Note the “remainder”

Until number ÷ *base* is 0

$53241 \div 16 = 3327, R = 9_{10} = 9_{16}$

$3327 \div 16 = 207, R = 15_{10} = F_{16}$

$207 \div 16 = 12, R = 15_{10} = F_{16}$

$12 \div 16 = 0, R = 12_{10} = C_{16}$

$53241_{10} = CFF9_{16}$

Base n to Decimal

Initialise sum to zero

Repeat for each digit

Multiply sum by *base*

Add digit to sum

$C_{16} = 12_{10} = (0 \times 16) + 12 = 12$

$F_{16} = 15_{10} = (12 \times 16) + 15 = 207$

$F_{16} = 15_{10} = (207 \times 16) + 15 = 3327$

$9_{16} = 9_{10} = (3327 \times 16) + 9 = 53241$

$CFF9_{16} = 53241_{10}$

Bin to / from Oct to / from Hex

- Binary to / from Octal

  Group bits into 3, use equivalence table

- Binary to / from Hexadecimal

  Group bits into 4, use equivalence table

- Octal to / from Hexadecimal

  Convert to Binary first

Storing Integers

- Binary Coded Decimal
- Sign and Magnitude
- Two's Complement

BCD: Binary Coded Decimal

- Group of 4-bits per decimal digit
- $0_{16} \cdots 9_{16} = 0_{10} \cdots 9_{10}$
  $A_{16}(10_{10}) \cdots F_{16}(15_{10})$ Invalid
- Decimal arithmetic is simple
- Binary arithmetic is not simple
- Storage requirement: 0 . . . 99
  vs 0 . . . 255

2004	0010 0000 0000 0100
− 1970	0001 1001 0111 0000
34	0000 0000 0011 0100

Sign and Magnitude

- MSB is Sign
  (0 = $+$ve, 1 = $-$ve)
- Remaining bits hold value (Magnitude)
- Range: $-2^{n-1} \cdots +2^{n-1}$
- $+0$ and -0
- Requires two hardware units: Adder and Subtractor

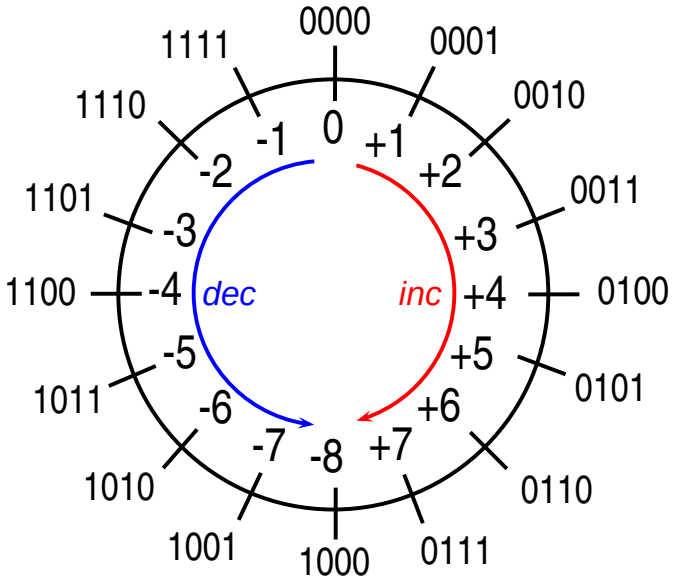
S 421		S 421	
0 000	$+0$	1 000	-0
0 001	$+1$	1 001	-1
0 010	$+2$	1 010	-2
0 011	$+3$	1 011	-3
0 100	$+4$	1 100	-4
0 101	$+5$	1 101	-5
0 110	$+6$	1 110	-6
0 111	$+7$	1 111	-7

Two's Complement

- MSB is -2^n
- A true complement
  $+X + (-X) = 0$
- Complement of Complement is itself: $-(-X) = X$
- Range: $-2^n \cdots +(2^n - 1)$
- There is a unique 0
- Subtractor *not* required

-8 421		-8 421	
0 000	0	1 000	-8
0 001	$+1$	1 001	-7
0 010	$+2$	1 010	-6
0 011	$+3$	1 011	-5
0 100	$+4$	1 100	-4
0 101	$+5$	1 101	-3
0 110	$+6$	1 110	-2
0 111	$+7$	1 111	-1

The Number Circle



0x9235 (1001 0010 0011 0101)

Binary Coded Decimal: 9235

Sign and Magnitude:

Sign: 1 (–ve)

Magnitude: $2^0 + 2^2 + 2^4 + 2^5 + 2^9 + 2^{12}$

$= 1 + 4 + 16 + 32 + 512 + 4096$

$= -4661$

Unsigned Integer: $2^0 + 2^2 + 2^4 + 2^5 + 2^9 + 2^{12} + 2^{15}$

$= 1 + 4 + 16 + 32 + 512 + 4096 + 32768$

$= 37429$

2's Compliment: $2^0 + 2^2 + 2^4 + 2^5 + 2^9 + 2^{12} - 2^{15}$

$= 1 + 4 + 16 + 32 + 512 + 4096 - 32768$

$= -28107$